

OFFICIAL OPERATIONS MANUAL

Hawkins Bullwhip Command Simulator

Decision-Support & Policy Simulation Guide for 5-Tier Supply Chain Stabilization

Institution:	Hawkins Cookers Limited (Plants: Thane, Hoshiarpur, Jaunpur)
Scope of Study:	12-SKU National Subset (39.8% Volume) 6 Channel Archetypes 104 Weeks
PRNG Seed:	31071959 (Mulberry32 Deterministic Seed)
Algorithm:	POUT Damped Replenishment (Proportional Order-Up-To Policy)

1. EXECUTIVE SUMMARY & THE 173% QUARTER-END SMOKING GUN

Founded in 1959, Hawkins Cookers Limited operates manufacturing facilities at Thane (Maharashtra), Hoshiarpur (Punjab), and Jaunpur (Uttar Pradesh). Over 105 cooker variants are delivered across India via 600+ retail dealers, 29 primary wholesale distributors, and 4 regional finished-goods depots.

CRITICAL EMPIRICAL FINDING — THE QUARTER-END COMMERCIAL DISCONNECT:

In the final weeks of each financial quarter, distributor primary orders surge by +173% above normal levels, while actual consumer kitchen offtake at retail counters runs 4% below average. This quarterly sales push induces a severe 15.77x Bullwhip Ratio (T2), destabilizing plant production schedules and tying up significant working capital in channel inventory.

2. THE 5-TIER SUPPLY CHAIN ARCHITECTURE

The simulator tracks weekly physical and information flows across 5 distinct tiers over 104 contiguous calendar weeks (two full fiscal cycles):

- Tier 0 (Consumer Kitchens):** Baseline retail offtake across India (~39,000 units/wk subset) featuring strong Diwali seasonality (2.1x peak). Bullwhip index = 1.00x.
- Tier 1 (Secondary Sales):** 600+ authorized retail dealer outlets holding ~2.9 weeks cover and pre-building stock 1.55x prior to the festive season. Bullwhip = ~2.81x.
- Tier 2 (Primary Orders):** Epicenter of volatility. 6 Channel Archetypes (representing 29 national distributors) ordering from Hawkins depots. AS-IS Bullwhip = 15.77x.
- Tier 3 (Central Depots):** Hawkins distribution warehouses buffering factory shipments with 4.2 weeks target cover and open-order backlog queueing. AS-IS Bullwhip = 46.03x.
- Tier 4 (Factory Production):** Assembly plants in Thane, Hoshiarpur, and Jaunpur operating on campaign run sizes (2.0 wks MOQ) with 1.95x surge capacity headroom. AS-IS Bullwhip = 20.17x.

3. MATHEMATICAL METHODOLOGY & CENTERED MA13 DETRENDING

Conventional variance calculations produce false alarms in seasonal industries because authentic consumer holiday surges (e.g., Diwali doubling demand) would mistakenly be flagged as "bullwhip".

Centered 13-Week Moving Average Detrending Formula:

$$\text{Detrended_t} = Y_t / \text{MA13}(Y)_t, \text{ where } \text{MA13}(Y)_t = (1/13) * \sum_{j=-6}^{+6} Y_{t+j}$$
$$\text{Bullwhip}(\text{Tier_k}) = \text{Var}(\text{Detrended Tier_k}) / \text{Var}(\text{Detrended Consumer Tier_0})$$

By dividing each weekly tier series by its centered 13-week moving average, long-term trends and festive seasonality are isolated, ensuring only artificial channel-induced whiplash is quantified.

4. PROPORTIONAL ORDER-UP-TO (POUT) REPLENISHMENT LAW

Under legacy order-up-to policies, distributors attempt to replace 100% of their stock deficit in a single week (alpha = 1.0), triggering violent upstream order amplification. The POUT policy applies proportional feedback damping:

$$O_t = Forecast_t + \alpha * (TargetStock - InventoryPosition_t)$$

Where alpha is the damping factor (0.10 to 1.00), spreading inventory adjustment over multiple weeks.

5. THE "SERVICE TRAP" PHENOMENON & DUAL-LEVER RESOLUTION

A central finding of this research suite is identifying the Service Trap: When distributors apply aggressive order damping (alpha = 0.35) without live retail Point-of-Sale (POS) visibility, bullwhip drops from 15.77x to 5.17x, BUT Consumer Service Level collapses from 97.5% down to 91.36%, causing severe stockouts during festive peaks.

Resolution: Order damping MUST be paired with POS Counter Telemetry (pout-35-pos). With shared retail data, Bullwhip drops to an optimal 2.60x while Customer Service Level reaches an exceptional 98.56%.

6. CALIBRATION BENCHMARKS MATRIX (6 CANONICAL PRESETS)

Each operational preset is calibrated against empirical simulation baselines. All values below represent the deterministic benchmark targets:

Preset ID	Policy	Alpha	POS %	Target BW (T2)	Service %	Fill Rate %	Operational Significance
as-is	ASIS	N/A	0%	15.77x	85.6%	80.0%	Legacy Push (+173% surge)
pout-100	POUT	1.00	0%	13.28x	98.3%	85.7%	Undamped POUT (No POS)
pout-60	POUT	0.60	0%	8.18x	97.5%	93.0%	Moderate Damping
pout-35	POUT	0.35	0%	5.17x	91.4%	96.6%	SERVICE TRAP (Diwali stockout)
pout-60-pos	POUT	0.60	100%	4.79x	98.2%	93.2%	Intermediate
pout-35-pos	POUT	0.35	100%	2.60x	98.6%	96.1%	GOLD STANDARD (Best balance)

7. STEP-BY-STEP OPERATOR WORKFLOW GUIDE

- Step 1: Baseline Audit:** In Executive Overview, select AS-IS preset. Observe 15.77x Bullwhip and quarterly order surges.
- Step 2: Simulate Damping:** Switch to pout-35 in Policy Sandbox. Bullwhip drops to 5.17x but warns of service degradation (91.4%).
- Step 3: Activate POS Telemetry:** Select pout-35-pos. Confirm Bullwhip drops to 2.60x while Consumer Service Level recovers to 98.6%.
- Step 4: Pipeline Inventory Review:** Review the Inventory Trends chart to observe the ~16% reduction in system inventory.
- Step 5: Verify Acceptance:** Navigate to Acceptance & Calibration Check to verify all 6 presets pass empirical tests within +/-15% tolerance.

8. PRNG SEEDING & VERIFICATION PROTOCOL

The simulator is 100% deterministic, utilizing the Mulberry32 algorithm initialized with seed 31071959 (Hawkins founding date 31-July-1959). Consumer variance models negative binomial noise (r = 8.0). To verify exact calibration, run the built-in Acceptance & Calibration Suite.